AWS CLOUD COMPUTING INTERNSHIP PROJECT REPORT

PROJECT TITLE: Implementation of Cross-VPC Peering with Multi-Tier Subnet Routing
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1. Project Overview & Objectives

In modern enterprise cloud environments, isolating workloads across distinct Virtual Private Clouds (VPCs) is standard practice for security, compliance, and organizational boundaries. However, backend services often require secure, high-speed, and private interconnectivity without exposing traffic to the public internet.

The primary objective of this project is to architect, configure, and validate a secure multi-VPC infrastructure on Amazon Web Services (AWS) utilizing:

1. **VPC Segmentation:** Defining custom IPv4 Classless Inter-Domain Routing (CIDR) blocks partitioned into dedicated public and private subnets.
2. **Internet Ingress/Egress Control:** Deploying and attaching an Internet Gateway (IGW) explicitly to the public subnet to host an administrative Bastion/Jump host.
3. **Private Subnet Isolation:** Keeping backend database/application subnets completely isolated with no route to the internet.
4. **AWS VPC Peering:** Creating an active network peering connection between two non-overlapping CIDRs (28.0.0.0/16 and 28.200.0.0/16).
5. **Bidirectional Route Tables & Security Groups:** Populating routing tables and stateful firewall rules to permit end-to-end ICMP and SSH communication across peering boundaries using private IP addresses.

2. Architectural Blueprint & Network Parameters

2.1 Network Topology



2.2 Resource Specification Matrix

Parameter	VPC 1 (Primary Workload)	VPC 2 (Target / Isolated)
VPC ID	vpc-07a66050da5464798	vpc-0fd21749c32e952e3
IPv4 CIDR Block	28.0.0.0/16	28.200.0.0/16
Public Subnet	VPC1-Public-Subnet (28.0.1.0/24)	None (Fully Private)
Private Subnet	VPC1-Private-Subnet (28.0.2.0/24)	VPC2-Private-Subnet (28.200.0.0/24)
Internet Gateway	igw-0ab6771d9c9d6f758	None
Route Table	rtb-0dc5536601fd48d4d (VPC1-Public-RT)	rtb-0f684d0effae6c8ed (VPC2-Private-RT)
Peering Connection ID	pcx-05ed1c5aca3c42e60	pcx-05ed1c5aca3c42e60
Compute Instance	Server-1-Public & Server-2-Private	Server-3-Private

3. Step-by-Step Technical Implementation

3.1 Virtual Private Clouds and Subnet Provisioning

- **VPC Creation:** Created VPC-1 (CIDR 28.0.0.0/16) and VPC-2 (CIDR 28.200.0.0/16) in region ap-southeast-2 ensuring non-overlapping address space.
- **Subnet Segmentation:** Partitioned VPC-1 into VPC1-Public-Subnet (28.0.1.0/24) and VPC1-Private-Subnet (28.0.2.0/24). Partitioned VPC-2 into VPC2-Private-Subnet (28.200.0.0/24).
- Enabled **Auto-assign public IPv4 address** on VPC1-Public-Subnet for external access.

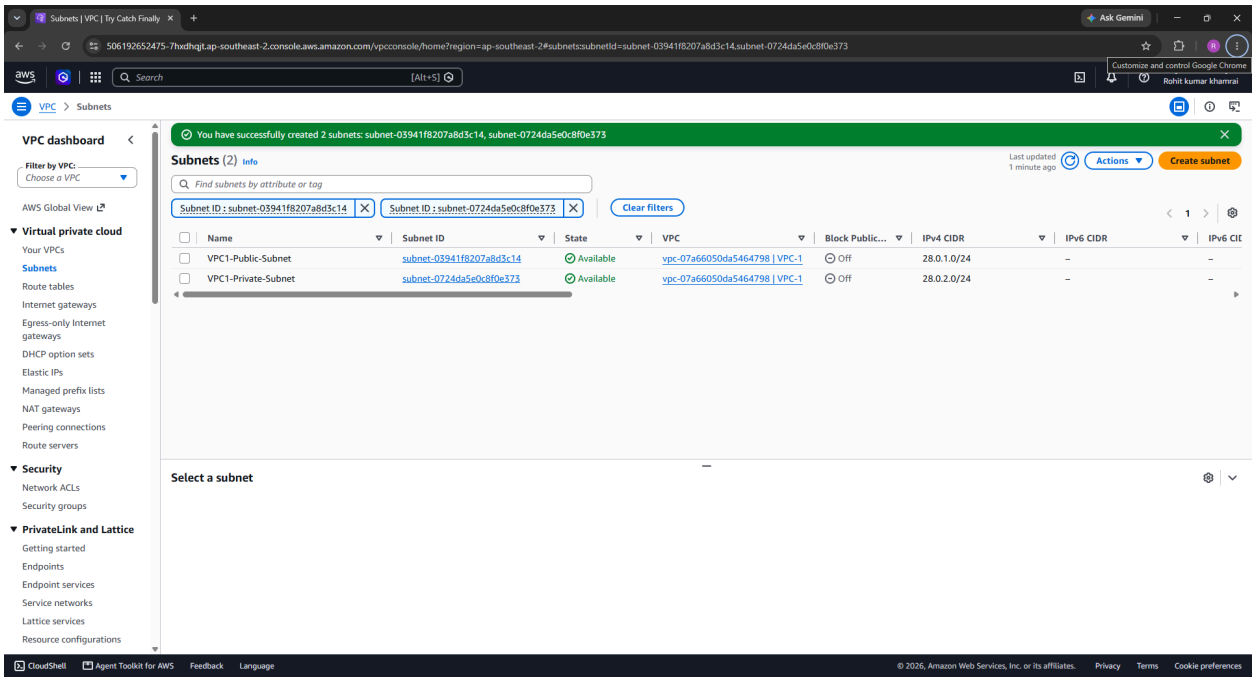


Figure 1: Subnet Configuration and Network Segmentation in VPC-1

3.2 Gateway Integration & Peering Connection Establishment

- **Internet Ingress:** Attached Internet Gateway igw-0ab6771d9c9d6f758 directly to VPC-1.
- **Peering Request:** Created peering connection VPC1-to-VPC2-Peering between VPC-1 and VPC-2.
- **Acceptance:** Accepted the connection request, changing its lifecycle status to **Active** (ID: pcx-05ed1c5aca3c42e60).

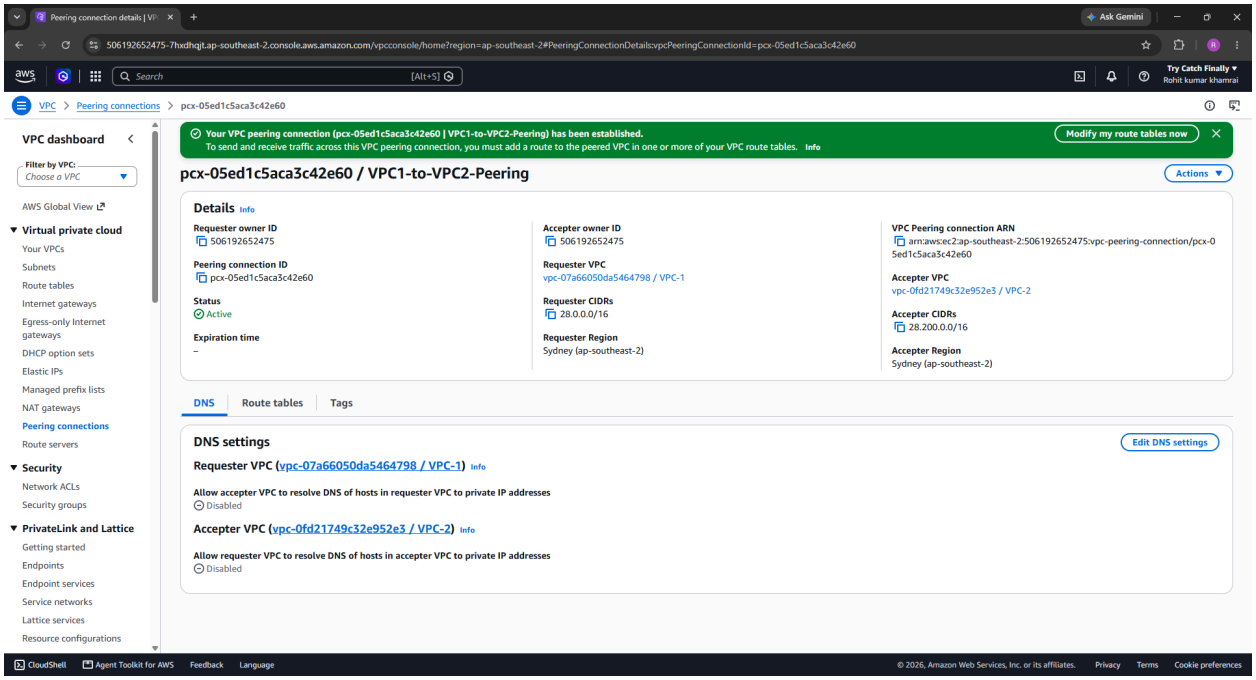


Figure 2: Active Cross-VPC Peering Connection Status

3.3 Routing Table Configuration

- **VPC-1 Public Route Table (rtb-0dc5536601fd48d4d):**
  - 28.0.0.0/16 -> local
  - 0.0.0.0/0 -> igw-0ab6771d9c9d6f758
  - 28.200.0.0/16 -> pcx-05ed1c5aca3c42e60
- **VPC-2 Private Route Table (rtb-0f684d0effae6c8ed):**
  - 28.200.0.0/16 -> local
  - 28.0.0.0/16 -> pcx-05ed1c5aca3c42e60

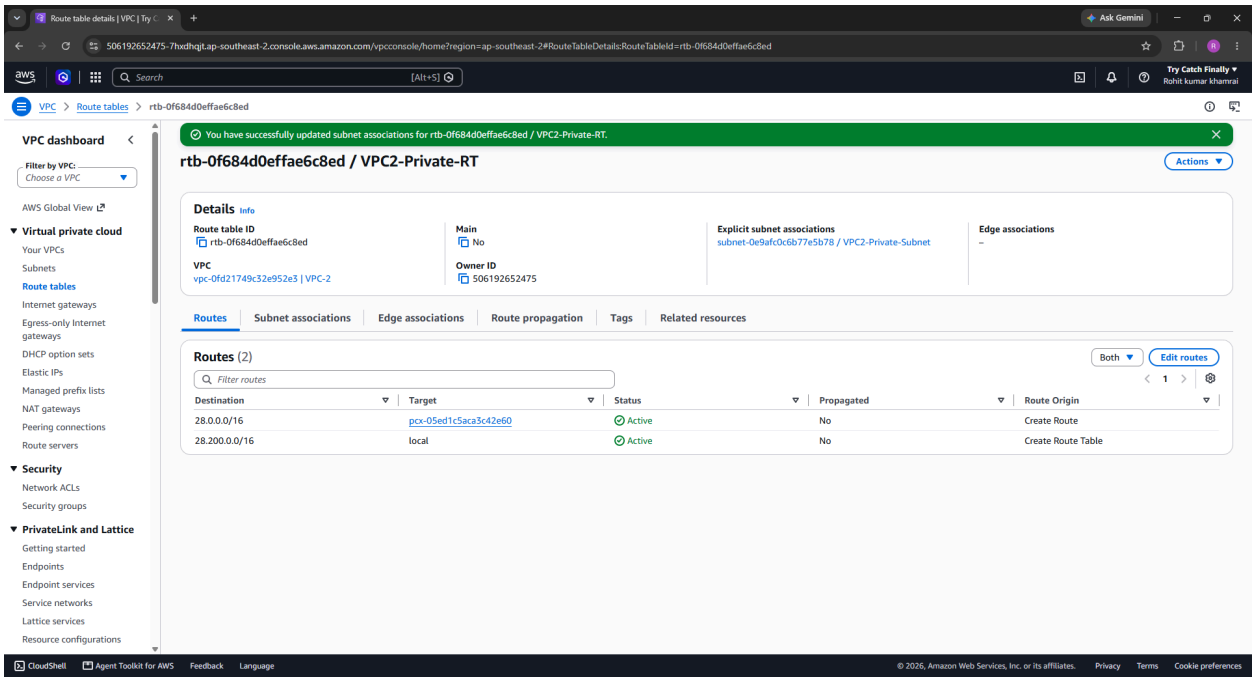


Figure 3: Route Table Configuration for Subnet Traffic Routing

3.4 Compute Instance Deployment & Stateful Firewalling

Provisioned three Amazon Linux 2023 t3.micro instances:

- **Server-1-Public (i-0fe724126cbcba02c):** Public IP 3.106.216.2, Private IP 28.0.1.60. Allowed SSH (Port 22) and ICMP IPv4 from anywhere.
- **Server-2-Private (i-066ea591ee9c1c720):** Private IP 28.0.2.225. Allowed SSH and ICMP from 28.0.0.0/16.
- **Server-3-Private (i-0c86e9a6d5ccbe63f):** Private IP 28.200.0.35. Allowed SSH and ICMP from 28.0.0.0/16 via VPC Peering.

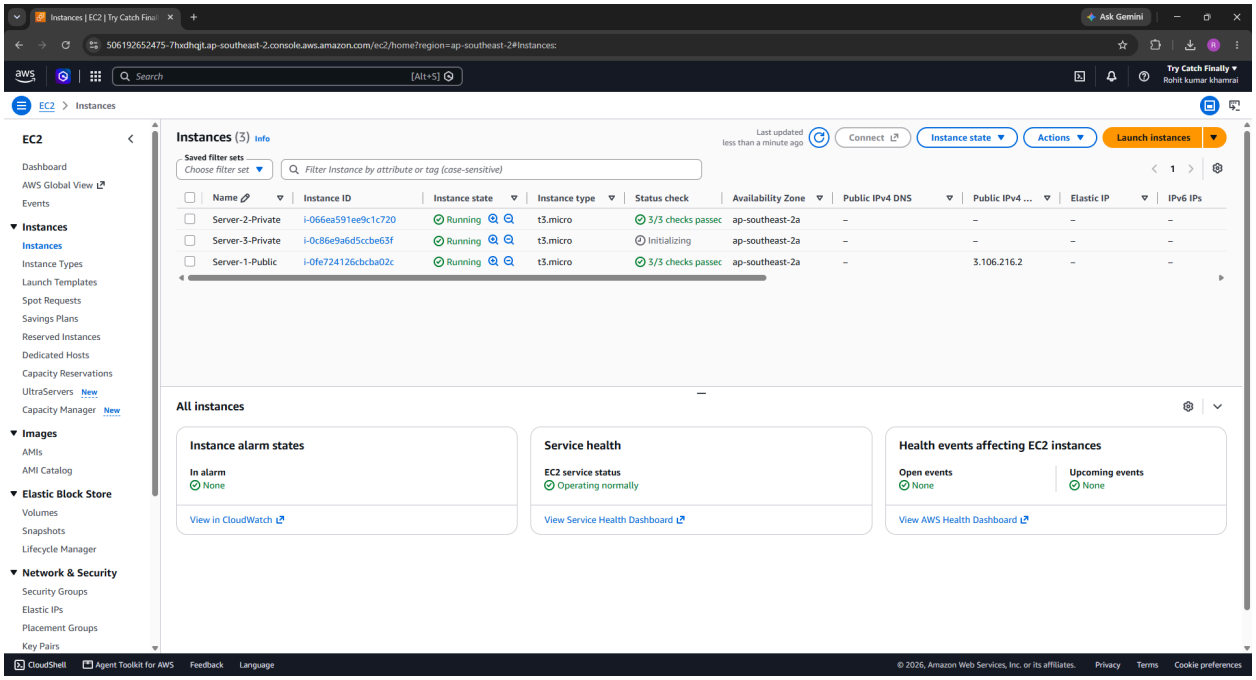


Figure 4: Deployed Compute Instances and Instance Status

4. Verification & Network Diagnostics Evidence

Connected to Server-1-Public via EC2 Instance Connect and executed diagnostic ICMP probes.

Test Case 1: Intra-VPC Public-to-Private Subnet Validation

```
[ec2-user@ip-28-0-1-60 ~]$ ping -c 4 28.0.2.225
PING 28.0.2.225 (28.0.2.225) 56(84) bytes of data.
64 bytes from 28.0.2.225: icmp_seq=1 ttl=127 time=0.410 ms
64 bytes from 28.0.2.225: icmp_seq=2 ttl=127 time=0.192 ms
64 bytes from 28.0.2.225: icmp_seq=3 ttl=127 time=0.184 ms
64 bytes from 28.0.2.225: icmp_seq=4 ttl=127 time=0.546 ms

--- 28.0.2.225 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3111ms
rtt min/avg/max/mdev = 0.184/0.333/0.546/0.152 ms
```

Result: PASSED (0% packet loss, 0.333 ms avg latency).

Test Case 2: Cross-VPC Peering Interconnectivity Validation

```
[ec2-user@ip-28-0-1-60 ~]$ ping -c 4 28.200.0.35
PING 28.200.0.35 (28.200.0.35) 56(84) bytes of data.
64 bytes from 28.200.0.35: icmp_seq=1 ttl=127 time=0.492 ms
64 bytes from 28.200.0.35: icmp_seq=2 ttl=127 time=0.170 ms
64 bytes from 28.200.0.35: icmp_seq=3 ttl=127 time=0.733 ms
64 bytes from 28.200.0.35: icmp_seq=4 ttl=127 time=0.188 ms

--- 28.200.0.35 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3086ms
rtt min/avg/max/mdev = 0.170/0.395/0.733/0.232 ms
```

Result: PASSED (0% packet loss, 0.395 ms avg latency). Bidirectional routing verified over private AWS backbone.

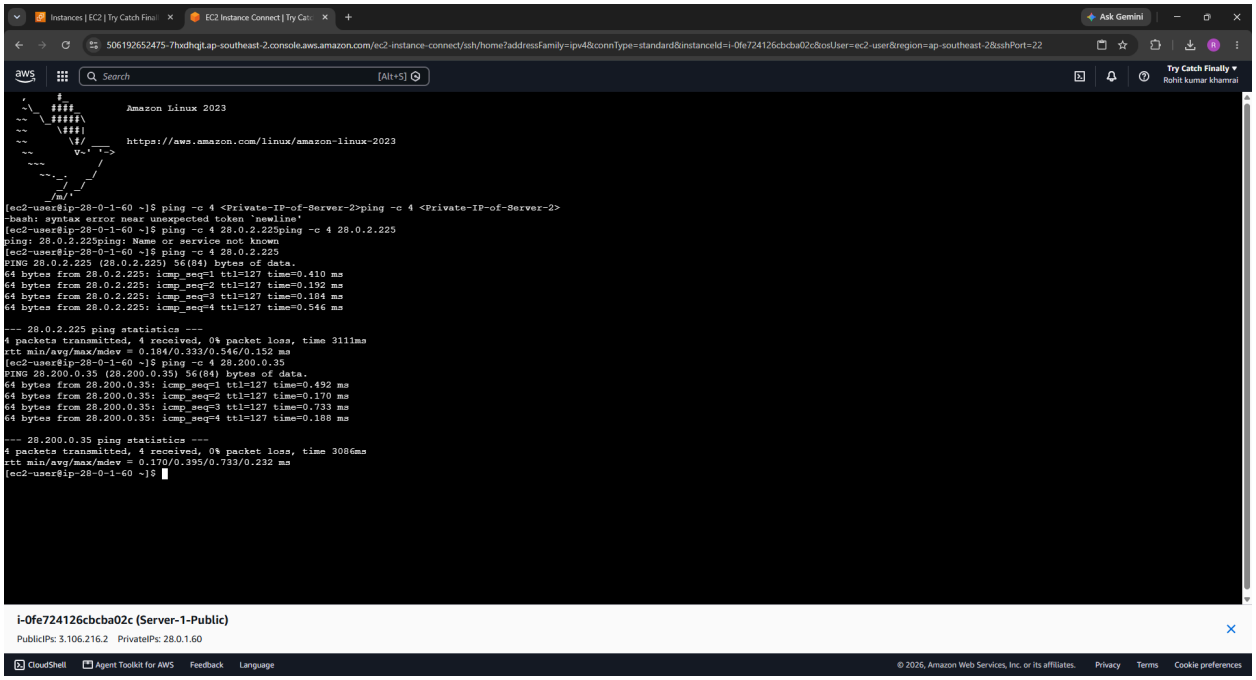


Figure 5: Network Diagnostics and Connectivity Verification Evidence

5. Security & Cost Optimization Measures

1. **Zero Internet Exposure for Backend Workloads:** Server-2-Private and Server-3-Private have no public IPs and cannot be addressed over the public internet.
2. **Least-Privilege Security Group Ingress:** Private instances restrict ICMP and management traffic strictly to internal CIDR ranges (28.0.0.0/16).
3. **Resource Decommissioning:** All three EC2 instances were cleanly terminated immediately following verification to prevent ongoing compute charges.

6. Conclusion

The multi-tier VPC architecture successfully validated private cross-network routing through an active AWS VPC Peering connection. All data packets between VPC-1 and VPC-2 remain fully confined within the Amazon private network fabric, meeting enterprise standards for isolation, performance, and security. By isolating backend compute workloads within private subnets and routing management traffic through a Bastion jump host, this implementation achieves a robust security posture while ensuring zero exposure of internal workloads to the public internet.

7. Deployed Resources Manifest

4. **VPC 1:** vpc-07a66050da5464798 (28.0.0.0/16)
5. **VPC 2:** vpc-0fd21749c32e952e3 (28.200.0.0/16)
6. **VPC 1 Public Subnet:** subnet-03941f8207a8d3c14 (28.0.1.0/24)
7. **VPC 1 Private Subnet:** subnet-0724da5e0c8f0e373 (28.0.2.0/24)
8. **VPC 2 Private Subnet:** subnet-0e9afc0c6b77e5b78 (28.200.0.0/24)
9. **Internet Gateway:** igw-0ab6771d9c9d6f758
10. **VPC Peering Connection:** pcx-05ed1c5aca3c42e60
11. **Route Tables:** rtb-0dc5536601fd48d4d (VPC 1) and rtb-0f684d0effae6c8ed (VPC 2)
12. **Compute Instances:** Server-1-Public (i-0fe724126cbcba02c, 28.0.1.60 / 3.106.216.2), Server-2-Private (i-066ea591ee9c1c720, 28.0.2.225), Server-3-Private (i-0c86e9a6d5ccbe63f, 28.200.0.35)
13. **Key Pair:** aws-internship-key.pem